

Choosing Configuration and Compilation Options



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TypeScript Compiler Essentials

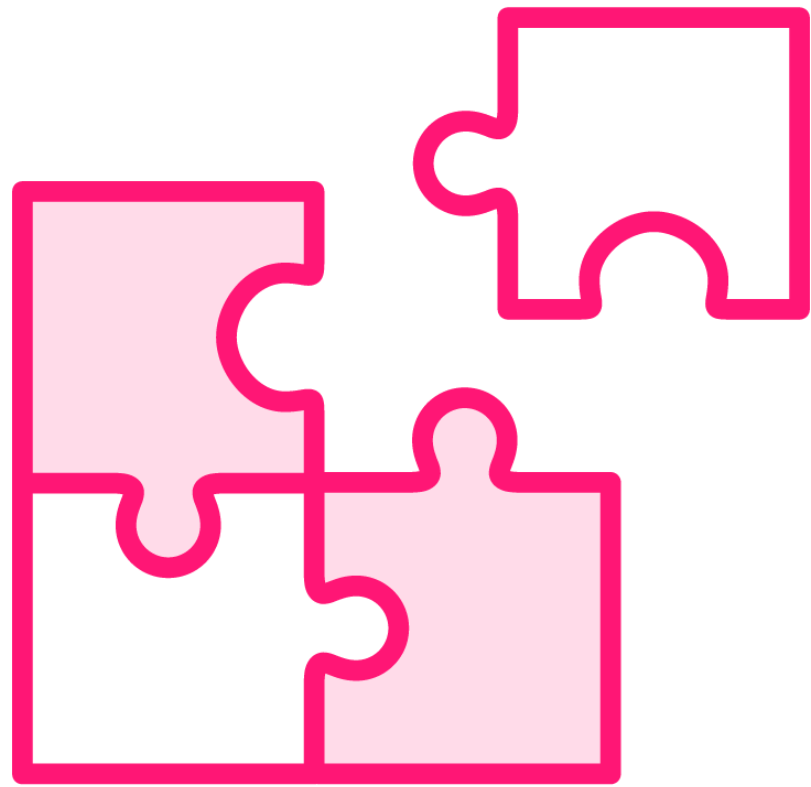
Here's what we know

"translate"

No execution
without transpilation



The Compilation Process



A good understanding of the process

Read the file, enforce the rules

And translate it into a *syntax tree*

A form independent of the language

ts2python - <https://pypi.org/project/ts2python>

<https://github.com/microsoft/TypeScript/wiki/Using-the-Compiler-API>

Static analysis and more rules to enforce





Demo: A Closer Look at Our Transpilation



Configuring Your Project with tsconfig Files



Some Go-tos in TSConfig



target



module



allowJs



removeComments



newline



strict

Try to always make it true

**Maybe set it to false while you get
stuff straightened out**

An umbrella option

**Turn on as many of the options as
you can at least**



forceConsistentCasingInFileNames

Casing in *imports*

“Person.ts” vs.
“person.ts”

Not much of a case
for false



Demo: Working with tsconfig Files



Using Type Declaration Files



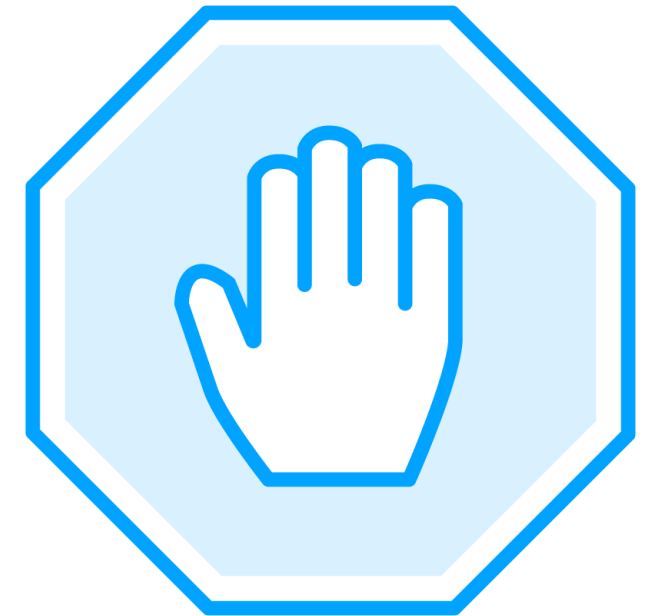
The Two Definitions



Runtime code

[1, 2, 3]

A type definition



**Third-party libraries
only guarantee
runtime code**



The Good News

Many packages
are including
type definitions

Any at runtime
(functionally)

Generating type
definitions is trivial
when you're working
in ts



The Better News

You can usually get it
somewhere else

[https://github.com/
DefinitelyTyped](https://github.com/DefinitelyTyped)

```
npm install -  
  save-dev  
  @types/react
```



Where That Will Get You

Pretty far

**A major library –
either supplied
or supported**

**A narrow window for
one with no support**



Demo: Inferring Types from Type Declaration Files



What's Going on Here

```
import { untypedCalculator } from "../untyped"
```

```
/* Yes, it is pointing at the untyped.js file or it is  
pointing at the untyped.d.ts file, depending on whether  
it's at runtime or design-time */
```

